



SHAPE OF INNOVATION

Design. Build. Inspire.
Innovative ideas take shape
with creativity and purpose



INNOVATIVE

Turning ideas
into smart
solutions



BUILD

Crafting with
precision and
passion



INSPIRE

Empowering the
future with
creativity

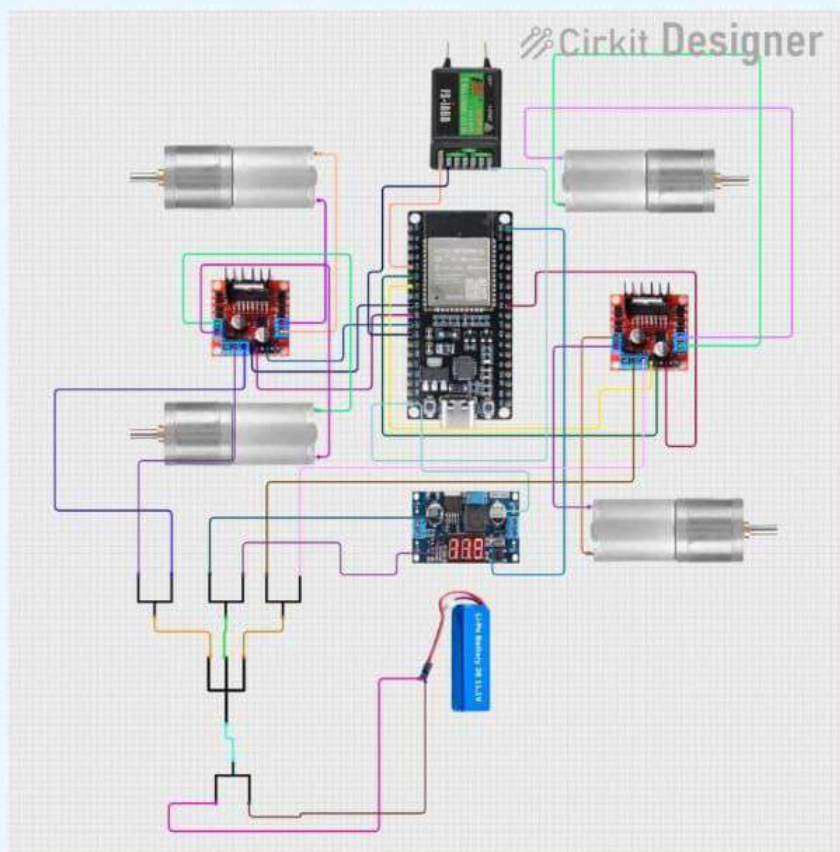


IN SEARCH OF
INNOVATION

BUILDNEO: A student-led research and development initiative at the Indian Maritime University, Visakhapatnam, focused on innovation and practical skill development in naval architecture. Founded by ambitious alumni, it thrives on a network of past and present students collaborating on working prototypes and research. The initiative features a dynamic leadership team, including roles in external communications, electronics, technical documentation, coding, CAD modeling, fabrication, additive manufacturing, and social media. By promoting project flexibility and creativity, BuildNEO prepares members for academic and industrial challenges, aiming to become a hub of talent and innovation in the maritime industry.

BOT BALL

This project emphasizes transforming theoretical concepts into practical, hands-on implementation. We developed a robust robotic architecture designed to complete complex, points-based tasks within a highly constrained and competitive environment. A collaborative, iterative approach was followed to ensure efficient mechanical integration and reliable system performance. Continuous testing and refinement played a key role in improving our coding structure and overall design stability. The time and effort invested by the team helped us overcome multiple build-phase challenges through innovative solutions. This presentation outlines our technical journey, the obstacles encountered, and the strategies that led to the successful deployment of our final Botball system.



FEATURES

The robot turns and changes direction based on differential steering. This principle states that for moving in any desired direction, the two motors on either side should act as a pair. For turning left, the left pair of motors (front and back) should rotate in the forward direction, while the right pair of motors rotate in the reverse direction simultaneously to achieve the desired turn. The same concept works vice versa for turning the robot to the right. This is the most basic application of differential steering.

ELECTRICAL COMPONENTS

- 25 GA 370 12V 625 RPM MOTORS
- ESP-32 (MICRO - CONTROLLER)
- BTS 7960 MOTOR DRIVERS
- DIGITAL BUCK CONVERTER
- 11.1V LiPo BATTERY
- XT-60 SPLITTERS
- FS-IA6B RECIEVER & TRANSMITTER

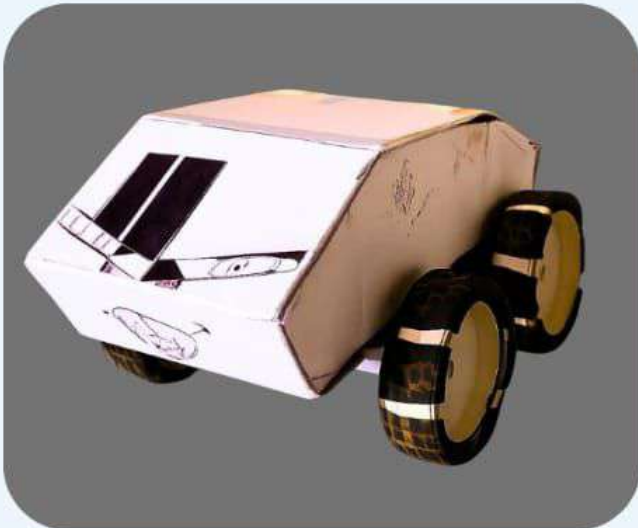


CODE - BLOCK & WIRING SEQUENCE

This code controls a differential drive robot using an RC receiver and two BTS7960 high-power motor drivers whereas the wiring sequence tells the Electrical Logic Functioning

WORKING OF THE CODE :

- **Signal Processing:** The `pulseIn()` function reads PWM signals from the RC receiver (steering and throttle). A safety check ensures that if the signal is lost or out of bounds, the motors stop immediately.
- **Mixing Logic:** The raw signals are mapped to a range of -255 to 255. The code uses arcade drive logic, where throttle and steering inputs are combined (Left = Throttle + Steer; Right = Throttle - Steer) to allow for fluid movement and turning.
- **Motor Control:** The `driveBTS7960` function translates these values into physical motion. By toggling the RPWM and LPWM pins, it controls both the direction and speed (via PWM) of each motor side.



WIRING SEQUENCE

The circuit has two parallel power lines one goes straight into the motor drivers so that the motors get max voltage for smooth functioning and other line goes to buck converter which steps down the which voltage to 5V , becoming the power line for the esp 32 termed as the logic line which sends signal and acts as a brain and nervous system of the botball . The esp 32 recieves commands from 2.4GHz-6 channel radio reciever using a reciever and a controller. The bot moves forward and backward and turns forward based on the principle of Differential steering